

- **Broca's area** It is present in brain and is related with speech, while Wernick's area of brain is related with understanding speech.
- **Hypothalamus** It deals with water balance in body, behavioural patterns of sex, sleep, stress, emotions, etc. It also regulates pituitary hormones and metabolism of fat, carbohydrate and water.

Midbrain

It is located between hypothalamus and pons of the hindbrain.

- It consists of two pairs of rounded swelling lobes called **corpora quadrigemina** (soild structures).
- It deals with visual analysis, auditory, etc.
- Midbrain of rabbit consists of four optic lobes.

Hindbrain

It consists of cerebellum, pons varolii and medulla oblongata.

- **Cerebellum** It is in the bottom part of the head and back of it. It controls the coordination of accurate movements and balancing. It is considered as small brain.
- **Medulla oblongata** It is the lowermost part of brain, which is connected to spinal cord. It deals with control of heartbeats, blood vessels, breathing, salivary secretion and most of reflex and involuntary (uncontrolled) movements.
- **Pons Varolii** It is an oral mass that lies above medulla oblongata. It interconnects two cerebellar hemispheres and also joins the medulla with higher brain centres.

Spinal Cord

It is a long cord-like structure present at the back, centrally located and well-protected by bony vertebral column.

- It gives out 31 pairs of spinal nerves in man.
- It conducts impulses to and from the brain.
- It is the centre for reflex actions and provides means of communication between spinal nerves and the brain.

Reflex Action

It is unconditional, inborn and unlearned reflex to a stimulus.

Examples Blinking of the eye when an object comes near to our eyes suddenly, rapid withdrawl of hand, while burned, sneezing, coughing, yawning, knee-jerk reflex, etc.

Acquired Reflex Action

It is also called as **conditioned reflex** and dependent on past experience, training and learning. These are under cerebral control during learning. First demonstrated by **Ivan Petrovitch Pavlov** in hungry dog.

e.g. Learning of dancing, cycling, swimming, singing and driving, etc.

Peripheral Nervous System

It is composed of 12 pairs of cranial nerves and 31 pairs of spinal nerves.

- Cranial nerves emerge from brain, while spinal nerves arise from spinal cord. Different number of cranial nerves are present in different organism.
e.g. 10 pairs of cranial nerves are present in fishes and amphibians.
 - Number of cranial nerves found in rabbit is 12 (same as man) but 37 spinal nerves are present in rabbit.
 - The first 10 pairs are common for frog and rabbit.

List of Cranial Nerves

<i>Cranial</i>	<i>Number</i>	<i>Function</i>	<i>Distribution</i>
Olfactory	I	Sensory	Nose
Optic	II	Sensory	Retina of eye
Occulomotor	III	Motor	4 muscles of eyeballs
Trochlear (smallest)	IV	Motor	Superior oblique muscle
Trigeminal (largest)	V	Mixed	
Ophthalmic	...		Snout
Maxillary	...		Upper jaw
Mandibular	...		Lower jaw
Abducens	VI	Motor	External rectus
Facial	VII	Mixed	Neck, pinna, tongue, lower jaw
Vestibulocochlear	VIII	Sensory	Internal ear
(auditory)			
Glossopharyngeal	IX	Mixed	Tongue, pharynx
Vagus	X	Mixed	Larynx, heart, lungs, stomach, etc.
Spinal accessory	XI	Motor	Neck
Hypoglossal	XII	Motor	Neck and tongue

Autonomous Nervous System

It was discovered by **Langley**. It is entirely motor and operates without conscious control.

Autonomic nervous system consists of two divisions:

- Sympathetic nervous system** This increases defence system of body against adverse conditions and active in stress conditions, e.g. pain, fear and anger.
- Parasympathetic nervous system** It provides relaxation, comfort, pleasure at the time of rest and helps in the restoration and conservation of energy.

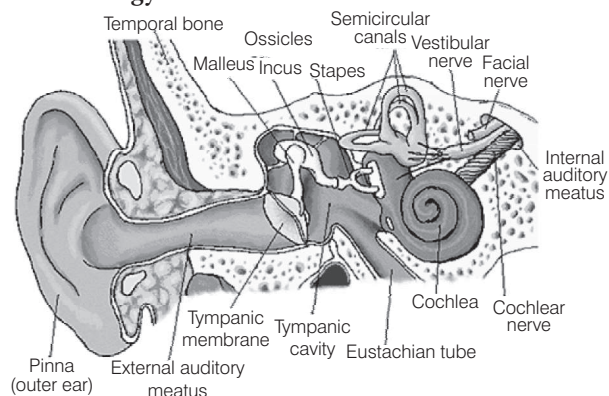
SENSE ORGANS

The sensory organs (receptors) enable us to detect all types of changes in the environment and send appropriate signals to the CNS, where all the inputs are processed and analysed signals are then sent to different centres of brain.

The different types of sense organs are as follows:

Human Ear

Ears are a pair of stato-acoustic organs meant for both sensory functions, i.e hearing and maintenance of body balance. The study of structure, function and disease of ear is called **otology**.



Structure of human ear

- These are located on sides of head.
- Hearing is controlled by auditory area of temporal lobe of cerebral cortex.
- The measuring unit of sound frequency is decibel. Human ear can listen the sound of 60-80 decibel.
- Human ear is sensitive to sound frequency 50-20000 cycles/sec and most sensitive to a frequency of 1000-3000 cycles/sec.
- Human ear is divided into three parts as:
- **External ear** (pinna + external auditory meatus), **middle ear** (tympanic cavity), **internal ear** (membranous labyrinth).

(i) External Ear

- The external organ consists of pinna and auditory canal.
- **Pinna** is designed to collect sound waves.
- **Auditory canal** contains few hairs and specialised sebaceous glands called **ceruminous glands**. These glands secrete cerumen (ear wax), it prevents foreign particles from entering the ear.

(ii) Middle Ear

- Tympanic membrane (ear drum) separates middle ear from external ear.
- The middle ear consist of three auditory ossicles. These are
 - (a) **Malleus** Outer and hammer-shaped.
 - (b) **Incus** Middle and oval-shaped.
 - (c) **Stapes** Inner and stirrup-shaped attached on the outer surface of fenestra ovalis membrane.
- The smallest bone in human body is stapes.
- The passage connecting the middle ear with pharynx is Eustachian tube.
- **Eustachian tube** equalises pressure in both sides (external and middle ear) or tympanic membrane.

(iii) Internal Ear

The inner ear consists of a labyrinth of fluid-filled chambers within the temporal lobe of the skull.

Labyrinth consists of two main divisions:

- (a) **Bony labyrinth** is filled with a fluid called **perilymph**, while membranous labyrinth contains endolymph.
- (b) **Membranous labyrinth** is concerned with both balancing and hearing.

- Internal ear is formed of three parts:
 - (a) Body proper (utricle and sacculus)
 - (b) Semicircular canals
 - (c) Cochlea
- Cochlea is small snail-shaped structure.
- It makes $2\frac{1}{2}$ turns in rabbit and $2\frac{2}{4}$ turns in man.
- Three chambers in cochlea are **scala vestibuli**, **scala media** and **scala tympani**.
- Scala media contains the organ of hearing named organ of Corti (discovered by **Alfonso Corti**).
- Defects of ear are otalgia earache (pain in ear) acute infection of middle ear, labyrinthine disease (malfunction of ear).

Human Eye

The organ of sight is a pair of eyes in human. Eyes are situated in deep bony cavities called **orbits** or **sockets** of the **skull**. These are the sensitive detectors of light. It is most sensitive to yellow colour.

- The study of structure, functions and diseases of eye is called **ophthalmology**.
- Eyeball is made basically of three concentric layers:
- **Outermost fibrous layer** composed of sclera and cornea.
- **Cornea** is thin transparent front part of sclera, which locks blood vessels, but is rich in nerve endings.
- **Sclera** outermost, tough and opaque part. It maintains the shape of eye.
- **Middle layer** consists of choroid, ciliary body and iris.
- **Choroid** chocolate coloured, pigmented (with melanin) present along innerside of sclera.
- **Iris** consists of circular sphincter and radial dilators. It adjusts the size of pupil.
- **Pupil** is the black hole in the centre of the iris, through it light enters the eyeball. It also controls the light entering into the eye.
- **Conjunctiva** Cornea is covered by a thin transparent layer called **conjunctiva**. It refracts the incident light rays to focus on the retina.
- **Lens** is a biconvex, transparent, circular, solid structure and is proteinaceous in nature.
- **Aqueous humor** is filled between cornea and lens.
- **Vitreous humor** is filled in space between the lens and the retina.

- **Retina** is innermost and incomplete layer. It consists of a nervous tissue layer and a pigmented layer. Image of object is formed on retina.
- Retina is composed of two types of cells, which are rod cells and cone cells. The image formed on it is real and inverted.
- **Rods** are longer, slender and filamentous, while cones are shorter and thicker. These are highly sensitive of dim light, contain a reddish-purple pigment called **rhodopsin**.
- **Cones** are sensitive to bright light, hence differentiate the colours (i.e. red, green, blue).
- **Yellow spot** is in the exact centre of the retina also called **macula lutea**. The fovea centralis is the area of sharpest vision due to high concentration of cones.
- **The blind spot** (optic disc) has no rods and cones cells, hence no image forms in this region.
- **Tapetum lucidum** increases the sensibility of vision. It is made up of zinc, cysteine and guanine.
- The eyes of carnivores like, e.g. cat, dog, lion, seal, etc., glow in night due to the tapetum lucidum.
- **Atropine** is a chemical used by doctors to dilate the pupil.
- **Night blindness** is caused by deficiency of rhodopsin in rod cells.
- Colour blindness is caused due to deficiency of cones.

The visual pigments for colour are:

- (a) Erythropsin (sensitive to red)
- (b) Chloropsin (sensitive to green)
- (c) Cyanopsin (sensitive to blue)

Diseases of Eyes

Some common diseases of eye are as follows:

- **Myopia** (short-sightedness) In this, the image of distant object being formed in front of retina. It is corrected by using concave lens.
- **Hypermetropia** (long or far-sightedness) Image is formed behind the retina, corrected by using convex lens. The image of nearer object becomes blurred.
- **Astigmatism** Due to irregular cornea or lens, corrected by using cylindrical lens.
- **Cataract** This is an eye disease generally occurs in older people. Lens becomes opaque and can be corrected by operation and using biconvex glasses.
- **Conjunctivitis** Inflammation of conjunctiva by bacteria.

Nose (Olfactory Organ)

Nose is a sense organ for smell or olfaction. The nose contains mucus coated olfactory (smell) receptors.

The nasal cavity is lined by olfactory epithelium lines.

Olfactory epithelium have three principal kinds of cells:

- (i) Olfactory bipolar neurons
- (ii) Columnar epithelial cells
- (iii) Bowman's gland

Tongue (Gustatory Organ)

Organs for gustation are taste buds, which are present mainly in the mucosa of taste papillae present on upper surface of tongue.

Taste buds are numerous on the tongue, but also found on the soft palate, pharynx and epiglottis. A papilla may contain few to many hundreds of taste buds. Taste buds are more numerous in children than in adults. Location of taste buds for basic tastes:

- **Sweet** On the tip of tongue.
- **Salty** On upper surface of anterior half.
- **Sour** Lateral side of tongue.
- **Bitter** Near the base of tongue.
- Bitter taste is evolved by many organic substances, i.e. quinine, morphine, caffeine and urea.

GLANDS SYSTEM

In the vertebrate body glands may be classified on the basis of the presence or absence of ducts. Glands are of two types:

(i) Endocrine Glands

The word endocrine derives from Greek words, 'endo' means within and 'crines' means to secrete. The endocrine system is the collection of glands that produce hormones that regulate metabolism, growth and development, tissue function, sexual function, reproduction, etc.

- The science dealing with the study of endocrine glands and their secretion is called **endocrinology**.
- These glands do not have ducts (tubes) to carry their secretion to the target organs.
- These are also called as **ductless glands**.
- Their secretion is called **hormone** or **internal secretion**.
- These glands secrete their secretion into the blood for their transportation to the sites of action, e.g. thyroid, pituitary, pancreas, hypothalamus, adrenal, etc.

(ii) Exocrine Glands

- These glands have ducts.
- These glands secrete their secretions in ducts to carry them to sites of action, e.g. sweat and oil glands of skin, salivary gland, liver, etc.

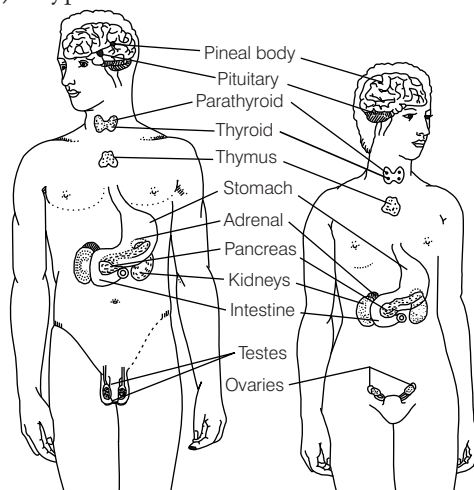
Hormones

- These are the chemical messengers that can regulate biological processes. They are steroids, proteins, peptides of amino acid derivatives.
- They can act as coenzymes.
- These are the organic compounds secreted in small amount by endocrine glands of body.
- Hormone was first discovered by **Bayless and Ernst H Starling** in 1903.
- Secretin was the first hormone to be discovered.

Endocrine Glands of Human

There are total nine endocrine glands in human beings. Most of them are common to male and female.

- | | |
|-------------------------------|---------------|
| (i) Pituitary (master gland) | (ii) Thyroid |
| (iii) Parathyroid | (iv) Thymus |
| (v) Adrenals (supra-adrenals) | (vi) Pancreas |
| (vii) Pineal gland | (viii) Gonads |
| (ix) Hypothalamus | |



Human endocrine glands

Pituitary Gland

- It is the smallest gland, which is about the size of a pea, but plays a major role in endocrine system.
- It is also called as the **master gland**.
- It is slightly larger in woman than in man.
- It is located in region of forebrain.

Thyroid Gland

- It is the largest endocrine gland.
- It is found on ventral and lateral sides of upper part of the trachea in the neck.
- It is brownish-red, shield-shaped, bilobed gland.

Hormones secreted by it are given in the table below:

Parts of pituitary	Hormones	Target organs	Actions/Effects
Anterior pituitary	Growth Hormone (GH)	All tissues	Normal body growth.
	Adrenocorticotropin Hormone (ACTH)	Adrenal cortex	Glucocorticoid secretion.
	Thyrotropin Stimulating Hormone (TSH)	Thyroid gland	Thyroxine secretion.
	Prolactin	Alveolar cells of mammary glands	Milk secretion.
	Follicle Stimulating Hormone (FSH)	Ovarian follicles	Growth of ovarian follicle and oestrogen secretion in females and spermatogenesis in male.
	Leuteinising Hormone (LH)	Testes or ovaries	Development of corpus luteum and progesterone secretion in females, testosterone secretion in males.
Intermediate pituitary	Melanocyte Stimulating Hormone (MSH)	Skin	Synthesis of melanin in the skin.
Posterior pituitary	Oxytocin (protein)	Uterus, mammary glands	Milk ejection, uterine contraction, contraction of smooth muscles.
	Vasopressin (ADH)	Kidneys and arteries	Hypertonic urine, absorption of water, arteriolar constriction.

Hormones secreted by it are given in the table below:

Hormones	Target organs	Actions/Effects
Calcitonin (CT)	Bones, kidney	Fall in blood calcium, excretion of Ca^{2+} .
Thyroxine	Heart, liver, kidney, skeletal muscles, mast cells.	Tissue metabolism, growth, IQ, differentiation of gonads, metamorphosis in amphibians.

Parathyroid Glands

- These are four in numbers and are embedded in thyroid glands.
- These are small glands, which are oval-shaped and yellow coloured and present in the human neck.
- Parathyroid gland produces parathyroid hormone.

Hormones	Target organs	Actions/Effects
Parathormone (PTH)	Bones, kidney	Rise in blood calcium level, phosphorus check of calcium secretion.

Pancreas

- It is a mixed gland with both exocrine and endocrine function and located within the curve of duodenum.
- Three kinds of cells are found in islets of Langerhans namely alpha, beta and delta cells.
 - (i) α -cells Larger and peripheral cells, produce glucagon hormone.
 - (ii) β -cells Central and smaller cells, produce insulin that a peptide hormone.
 - (iii) δ -cells Middle cells produce somatostatin hormone.

Hormones secreted by the pancreas are given in the table below:

Hormones	Target organs	Actions/Effects
Insulin	All cells	Decreases blood sugar level, stimulates glycogenesis, increases stored protein in tissue. Hypersecretion results into diabetes mellitus.
Glucagon	Liver	Rise in blood sugar.

Adrenal Glands

- Our body has paired endocrine glands, located superior to kidneys. It is structurally and functionally divided into adrenal cortex and adrenal medulla.
- It is also called as **4S gland** because of its four functions including salt retention, sugar metabolism, as a source of energy and as a sex organ.

Hormones secreted by the adrenal gland are given in table below:

Hormones	Target organs	Actions/Effects
Glucocorticoids	Tissues, mast cells	Carbohydrate, fat and protein metabolism.
Mineral corticoids	Kidney	Sodium and potassium metabolism, water absorption.
Sex corticoids	Body cells	External sex characters.
Adrenaline and noradrenaline	Mast cells	Heartbeat, blood pressure rise, contraction and relaxation of muscles.

Pineal Gland

- It is a small gland in the brain derived from the embryonic ectoderm. It secretes hormones in duct response to nervous activity.
- In man, it starts to degenerate at about age of 7 years.
- In adult, it is largely a fibrous tissue.

Hormone secreted by pineal gland is given below:

Hormone	Target organ	Action/Effect
Melatonin	Melanophores	Dispersion of melanin.

Thymus Gland

- It is a lymphoid gland that plays an important role in the development of immune system.
- It is situated in front of the heart.
- Thymus is active in young ones, but gradually becomes inconspicuous after sexual maturity.
- It arises from the endoderm of the embryo and consists of peripheral cortex and central medulla.

Hormone secreted by thymus gland is given below:

Hormone	Target organ	Action/Effect
Thymosin	—	Immune system of body produces lymphocytes.

Hypothalamus

The complete endocrine system works more or less under the influence of hypothalamus.

- It is located in the floor of forebrain.
- Neurohormones were first discovered by **Guillemin** and **Schally**.

Hormones secreted by hypothalamus are given below:

Hormones	Target organs	Actions/Effects
Thyrotropin releasing hormone	Thyroid, adrenal gland	Thyrotropin secretion, Corticotropin secretion
Corticotropin releasing hormone	Pituitary	Secretion of pituitary gonadotropin
Somatostatin	Pituitary	Initiation of growth hormone secretion

Gonads

- The main function of gonads is to produce gametes.
- They also secrete sex hormones.
- Sex hormones are mostly steroids.
- The testes in male and ovaries in females are gonads.

Hormone secreted by gonads are given below:

Endocrine gland	Hormones	Target organs	Actions/Effects
Testes	Testosterones	Male reproductive organ	Male secondary sex organs and external male characters.
Ovaries	Oestrogen	Female reproductive organ	Female secondary sex organ and external female characters.
Corpus luteum (ovaries)	Progesterone	Uterus, mammary glands	Pregnancy changes in female sex organs.

Apart from these, hormones are also secreted from heart kidney and gastro-intestinal tract.

Organ	Hormone	Role
Heart	Atrial natriuretic factor	Dilation of blood vessels
Kidney	Erythropoietin	Erythropoiesis
Gastro-intestinal tract	Gastrin, Secretin, Cholecystokinin	Secretion of HCl Action pancreas for the secretion of water acts on gall bladder and pancreas for bile juice secretion.

REPRODUCTION IN ANIMALS

The process by which new individuals are produced from their parents is called **reproduction**. Reproduction is of two kinds:

